

S. PRAVEEN

AI-Native Scientific Product Builder · Chemistry Researcher · Research Technology Creator

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PROFESSIONAL SUMMARY

M.Sc. Chemistry graduate and AI-native product builder who transforms scientific expertise into functional, deployed software tools — without traditional hand-coding. Combines graduate-level knowledge in electrochemistry and nanomaterials with AI-assisted development workflows (Claude AI, ChatGPT, prompt engineering, iterative prototyping) to conceive, design, and ship scientific applications. Proven across five deployed products serving chemistry researchers, NEET/JEE students, and medical education learners. Brings a unique blend of domain credibility, product thinking, and AI-native execution to roles at the intersection of science, software, and research technology.

CORE COMPETENCIES

Scientific Domain Expertise · AI-Assisted Application Development · Prompt Engineering & Specification Writing · Product Requirement Definition · Scientific Workflow Design · User-Centred Problem Framing · Iterative Feature Refinement · Scientific Accuracy Validation · Research Literature Synthesis · Science Communication & Curriculum Design · Deployment & Product Ownership

AI-NATIVE PRODUCT DEVELOPMENT

ChemSphere 3D — Deployed Scientific Visualization Platform

2026

praveens-chem.github.io/ChemSphere · AI-assisted development via Claude AI & ChatGPT

- ▶ Identified a clear gap in chemistry education: NEET/JEE students lack access to interactive 3D molecular visualization; defined product requirements, user flows, and UX specifications before a single line of code was produced.
- ▶ Directed AI-assisted development using prompt engineering and iterative refinement to implement a mobile-responsive molecular viewer rendering real-time 3D structures — including Buckminsterfullerene and complex organic compounds.
- ▶ Designed the full scientific UI/UX hierarchy (atom legend, compound search, contextual labels) optimized for mobile screens; validated all chemistry content for scientific accuracy.
- ▶ Deployed via GitHub Pages with zero-infrastructure cost; owns ongoing maintenance, feature iteration, and user experience optimization.
- ▶ AI tools used: Claude AI, ChatGPT · Methodology: prompt engineering, vibe coding, iterative testing, AI-assisted debugging.

NanoLit AI — Deployed Research Productivity Tool

2026

praveens-chem.github.io/NanoLit · AI-assisted development via Claude AI & ChatGPT

- ▶ Conceived the product from direct personal pain: experienced literature overload, jargon density, and slow synthesis workflows as a postgraduate researcher and CSIR-CECRI project associate.
- ▶ Designed an AI-assisted scientific literature workflow platform that simplifies, annotates, and surfaces key findings from papers in materials science and electrochemistry; authored full product specification and workflow architecture.
- ▶ Directed AI-native application development end-to-end — from prompt-driven prototyping through feature refinement, UX polish, and branded deployment; packaged with a modular interface and clean loading state.
- ▶ Validated the research workflow logic against real domain knowledge from own nanomaterial synthesis and characterisation experience (XRD, XPS, CV).

- ▶ AI tools used: Claude AI, ChatGPT · Methodology: prompt engineering, workflow design, iterative refinement, AI-assisted debugging.

MOFFit — Deployed Materials Informatics Workflow Tool

2026

praveens-chem.github.io/MOFFIT · AI-assisted development via Claude AI & ChatGPT

- ▶ Built an AI-assisted Metal-Organic Framework (MOF) visualization and exploration platform enabling users to explore porous crystalline structures used in gas storage, catalysis, and chemical sensing applications.
- ▶ Applied domain expertise from own nanomaterial synthesis research to define scientifically grounded MOF structure logic and validation criteria — translating graduate-level materials science into an accessible interactive tool.
- ▶ Owned full product lifecycle: problem definition, workflow design, prompt-driven development, feature iteration, scientific accuracy validation, and deployment via GitHub Pages.

ChemSphere Formula Hub — Deployed Chemistry Reference Platform

2026

praveens-chem.github.io/Chemsphere-Formula-Hub · AI-assisted development via Claude AI & ChatGPT

- ▶ Identified the need for a fast, reliable, mobile-friendly chemistry formula reference tool for NEET/JEE students and undergraduate learners; defined the information architecture and content scope before development.
- ▶ Directed AI-assisted development of a comprehensive interactive platform covering chemical formulas, reaction types, and key chemistry concepts; validated all content for scientific accuracy using graduate-level domain knowledge.
- ▶ AI tools used: Claude AI, ChatGPT · Methodology: prompt engineering, iterative refinement, AI-assisted debugging, GitHub Pages deployment.

AnatomySphere — Deployed Human Anatomy Visualization Platform

2026

praveens-chem.github.io/AnatomySphere · AI-assisted development via Claude AI & ChatGPT

- ▶ Identified a gap in accessible interactive anatomy education for NEET/medical students; designed the product scope, visualization structure, and content taxonomy covering human body systems and anatomical structures.
- ▶ Directed AI-assisted development of an interactive anatomy visualization platform; applied scientific accuracy validation and UX optimisation to ensure content meets the standards expected by medical and biology students.
- ▶ AI tools used: Claude AI, ChatGPT · Methodology: prompt engineering, iterative prototyping, AI-assisted debugging, GitHub Pages deployment.

RESEARCH EXPERIENCE

Research Associate (Project) — CSIR-CECRI, Madras Unit — Chennai

2019 – 2020

- ▶ Designed and characterized electrocatalysts for Proton Exchange Membrane (PEM) fuel cell applications at India's premier electrochemical research institution, under senior scientist supervision.
- ▶ Conducted electrochemical performance evaluation using Cyclic Voltammetry (CV) and Linear Sweep Voltammetry (LSV); synthesized and characterised nanomaterials via XRD, NMR, and XPS.
- ▶ Produced a structured project report on catalyst efficiency — developing the scientific communication and structured documentation skills directly applied to today's product specification and workflow design work.

PROFESSIONAL EXPERIENCE

TGT Chemistry & Academic Coordinator & Student Counsellor — Sanskar The Gurukul School, Chhattisgarh

Jan 2026 – Present

- ▶ Teach Chemistry to CBSE Classes 9–12 as TGT Chemistry; design lesson plans, lab demonstrations, and concept-driven learning materials that make complex chemical topics accessible and exam-ready for students.

- ▶ Coordinate CBSE curriculum delivery and faculty assignments as Academic Coordinator; design supplementary STEM learning modules applying systems-level instructional design thinking to academic planning.
- ▶ Counsel students on academic planning, subject selection, and career pathways in science; communicate complex scientific concepts clearly — skills directly transferable to user-centred product design and science communication.

Freelance Science Content Creator — Remote / Jagdalpur

2021 – Present

- ▶ Produce chemistry-focused digital content, study materials, and SEO-optimised educational articles for coaching centres and institutions using ChemDraw and AI-assisted writing tools.
- ▶ Demonstrates consistent ability to translate graduate-level scientific concepts into accessible, audience-specific communication — a core competency for scientific product ownership.

EDUCATION

M.Sc. Chemistry — Sathyabama Institute of Science & Technology

2020 · CGPA 9.3 / 10

B.Sc. Chemistry — Alva's Degree College, Mangalore University

2018 · 73.4%

TECHNICAL & SCIENTIFIC SKILLS

Scientific Domains Electrochemistry · Nanomaterial Synthesis · PEM Fuel Cell Electrocatalysis · Materials Characterisation · MOF Chemistry

Instrumentation XRD · NMR · XPS · Cyclic Voltammetry (CV) · Linear Sweep Voltammetry (LSV) · Chronoamperometry

AI-Native Development Claude AI · ChatGPT · Prompt Engineering · Specification Writing · Vibe Coding · Iterative Prototyping · AI-Assisted Debugging · Workflow Design · GitHub Pages Deployment

Product Skills Product Requirement Definition · User Problem Framing · Feature Prioritisation · UX Optimisation · Scientific Accuracy Validation · Product Ownership

Scientific Communication Science Writing · Curriculum Design · Technical Documentation · ChemDraw · MS Office Suite

Languages English · Hindi · Tamil · Kannada

CERTIFICATIONS & WORKSHOPS

- ▶ Internship Certificate — Introduction to Fuel Cells, CSIR-CECRI, Chennai
- ▶ Workshop — Maintenance & Calibration of Analytical Instruments, St. Aloysius College, Mangalore

A NOTE ON DEVELOPMENT METHODOLOGY

The applications listed above were not traditionally hand-coded from scratch. They were conceived, specified, directed, tested, and deployed using AI-native development workflows — including Claude AI, ChatGPT, prompt engineering, iterative prototyping, AI-assisted debugging, and vibe coding methodologies. Product ownership, scientific domain expertise, workflow design, user problem framing, and accuracy validation are the core competencies applied throughout. This reflects an emerging and legitimate mode of scientific product creation.